

What is an Active Directory (AD) Server?

An **Active Directory (AD) server** is a **Windows Server** that runs **Active Directory Domain Services (AD DS)**. It acts as a **centralized system** that manages **users, computers, permissions, and security policies** across a network.

In simple terms: It's like a **school principal's office** where student (user) info, attendance (logins), rules (GPOs), and access control are all maintained centrally.

The Server That Hosts AD is Called:

Domain Controller (DC)

What Active Directory Does:

Function	Description
Authentication	Validates usernames and passwords
Authorization	Grants access to resources (folders, printers, apps)
User Management	Add/delete/modify users, groups, and computers
Group Policy Control	Apply rules (GPOs) to users or systems across the organization
Directory Service	Stores info about users, computers, OUs, and network resources

How AD Server Works (Step-by-Step)

1. Join Client to Domain

- A Windows computer is joined to the domain:
 - Example: student1@codingseekho.local

2. User Logs In

- User enters login info at their PC
- Login request goes to **Domain Controller**

3. Authentication with AD

- The **AD server checks credentials**
 - Uses **Kerberos Protocol**
- If correct, a **login token** is given to the user

4. Access Resources

- User tries to open shared folder or printer
- AD checks if user has permission
- If **authorized**, access is allowed

Components of AD Server

Component	Description
Domain	Logical group of computers and users
OU (Organizational Unit)	Container for grouping users/computers
Group Policy (GPO)	Set of rules that apply across domain members
Schema	Defines data structure (user fields, attributes)
Global Catalog	Central database of searchable AD info

Real-World Example:

Task	Done by AD?	Example
Login to any company PC	☒ Yes	user@company.local
Automatically map a network drive	☒ Yes	GPO assigns Z:\ as shared folder
Prevent USB usage on PCs	☒ Yes	GPO rule applied to all systems
Reset user password	☒ Yes	Admin uses AD to reset it remotely